

RIYAN SUGIARTO

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineer with over 4 years of experience in backend development, specializing in Java, Spring Boot, and microservices architecture. Proven expertise in building scalable, high-performance applications and integrating cloud-based solutions. Strong problem-solving skills with a keen interest in blockchain technology and Web3 development. Adept at optimizing database performance, implementing API security best practices, and building event-driven architectures using Kafka.

Extensive experience in designing and implementing robust microservices architectures, including API gateway management, distributed tracing, circuit breakers, and service discovery. Highly skilled in system design, handling complex distributed systems, and ensuring scalability and fault tolerance. Passionate about adopting the latest technologies to drive innovation and efficiency in software engineering.

PROFESSIONAL EXPERIENCE

PT. ASURANSI KREDIT INDONESIA

JAKARTA, ID

Software Developer

2020-2022

Software Engineer specializing in backend development with extensive experience in designing, developing, and maintaining scalable systems in the financial and insurance sectors. Passionate about optimizing performance, ensuring security, and enhancing business processes through robust software solutions.

Experiences :

- Develop new features and improve existing platforms to support financial operations and product development.
- Manage and build new data pipelines to support financial reporting and reconciliation.
- Take ownership of new project development, ensuring quality and efficiency from inception to deployment.
- Participate in the full software development lifecycle, from designing solutions to production release.
- Conduct comprehensive software testing, including unit testing, integration testing, system testing, and user acceptance testing.
- Create and customize reports using JasperReports for financial and operational data analysis.

Selected Project Experience

- **Pemulihan Ekonomi Nasional (PEN)**
A government initiative to mitigate the economic impact of COVID-19 by supporting informal sectors and maintaining economic stability. Contributed to developing digital solutions that streamlined financial assistance distribution.
- **Agunan Bank Online**
A digital platform designed to simplify and accelerate the process of bank collateral insurance applications. The platform enables Askrindo partner banks to apply for bank collateral insurance, ensuring faster policy issuance. Coverage includes up to Rp. 10 billion for class I construction and Rp. 5 billion for class II construction.
- **Surety Bond Online**
Developed a system that provides guarantees to employers (Obligees) against defaults arising from non-fulfillment of obligations by the principal for construction and non-construction projects, ensuring contractual compliance and financial security.

Job Desk & Responsibility

Backend Development & System Architecture

- Design and develop scalable backend systems using Spring Boot and Java.
- Optimize database queries and implement indexing strategies in PostgreSQL and Microsoft SQL Server to enhance performance.
- Implement caching strategies using Redis to improve application responsiveness.

Deployment & Server Management

- Automate deployment processes using shell scripts and CI/CD pipelines for faster and more reliable releases.
- Configure and manage Nginx as a reverse proxy for backend services on on-premise servers.
- Implement and maintain security best practices, including SSL/TLS encryption, firewall configurations, and server hardening.

System Integration, API Development and Monitoring

- Develop and integrate RESTful APIs for seamless communication between microservices.

- Ensure API security using JWT authentication and OAuth2 protocols.
- Conduct load testing and stress testing to ensure the system can handle high traffic efficiently.
- Profile and optimize memory usage, CPU performance, and database queries to reduce latency.

Collaboration & Teamwork

- Work closely with front-end developers, system analyst and business stakeholders to align technical solutions with business needs.
- Participate in code reviews, pair programming, and technical discussions to improve code quality and maintainability.

Stack Highlights:

- **Languages** : Java
- **Frameworks** : Spring Framework and Spring Boot
- **Databases** : Postgresql, Microsoft SQL Server
- **DevOps & Deployment**: Nginx, Shell Scripting, Linux Scripting
- **Security**: SSL/TLS, OAuth2, JWT Authentication
- **Reporting Tools**: JasperReports, JRXML Templates

PT. XL AXIATA

JAKARTA, ID

API Developer

2022-2023

Software Engineer specializing in backend development and API management for one of Indonesia's leading telecommunications companies. Experienced in microservices architecture, API integration, and large-scale system migration.

Experiences :

- Lead the migration of SOAR (TIBCO) to a Microservices Architecture using Spring Boot, improving system scalability and maintainability.
- Develop and maintain RESTful APIs and backend services for high-traffic applications.
- Participate in full-cycle software development, including requirement analysis, system design, coding, testing, deployment, and monitoring.
- Support multiple levels of software testing, including unit testing, integration testing, system testing, and user acceptance testing (UAT).
- Manage APIs in the API Gateway, ensuring security, scalability, and efficient partner integration.
- Write and maintain technical documentation for system architecture, API specifications, and operational guidelines.

Selected Project Experience

- **info-paket**
Retrieves and processes real-time package subscription details for mobile users by integrating with the billing, subscription, and customer management systems, ensuring up-to-date package information. This service communicates with internal APIs and caching layers to optimize response times and reduce database load while providing accurate package status, including active, expired, and upcoming subscriptions.
- **info-quota**
Provides real-time quota usage tracking and analytics for subscribers by aggregating data from various network traffic monitoring, billing, and provisioning systems. The service ensures high availability and low-latency responses by utilizing distributed caching, event-driven processing, and data streaming technologies to deliver near-instant quota updates. Additionally, it supports threshold-based alerts for proactive quota management, helping users avoid service disruptions due to exhausted quotas.
- **incident-notification**
Manages automated SMS notifications for service incidents affecting XL Home (WiFi service) by integrating with network monitoring systems, incident management platforms, and SMS gateways. When a disruption or anomaly is detected, the service triggers real-time alerts to notify customers and support teams about ongoing issues, estimated resolution times, and necessary troubleshooting steps. It utilizes event-driven architectures, message queues (Solace), and RESTful APIs to ensure reliable and timely delivery of notifications, minimizing service impact and enhancing customer experience.
- **fulfillment-policy**
Implements and enforces business logic and validation rules for service fulfillment requests, ensuring that transactions comply with company policies, regulatory requirements, and customer eligibility criteria. This service acts as an intermediary between frontend applications and backend provisioning systems, orchestrating workflows for new

activations, upgrades, and plan modifications. It utilizes microservices architecture, message queues (RabbitMQ/Kafka), and API gateway policies to facilitate secure and efficient fulfillment operations.

- **fulfillment-subscriber-retry**

Enhances service reliability and fault tolerance by implementing intelligent retry mechanisms for failed fulfillment requests due to transient errors, network issues, or backend failures. The service uses asynchronous job queues, distributed task scheduling, and exponential backoff strategies to ensure successful processing of transactions while minimizing system overload. It supports automatic rollback and compensation logic in case of partially completed transactions, ensuring consistency across dependent services.

- **bulk-imei-registration**

Provides a high-performance batch processing service that enables telecom operators and regulatory bodies to register multiple device IMEIs simultaneously. This service supports large-scale data ingestion and validation pipelines, ensuring compliance with government regulations and fraud prevention policies. It utilizes parallel processing, database partitioning, and distributed computing frameworks (Solace) to efficiently handle bulk requests. Additionally, it integrates with device management platforms and global IMEI databases to verify device authenticity and track unauthorized devices.

Job Desk & Responsibility

Backend Development & System Architecture

- Design, develop, and deploy RESTful APIs using Spring Boot and TIBCO BW 5 / TIBCO BWCE.
- Implement asynchronous messaging using Solace for event-driven architectures.
- Develop high-performance, scalable microservices using Spring Cloud and containerized environments.

Deployment & Server Management

- Lead the migration of TIBCO SOAR services to a Spring Boot-based microservices architecture.
- Optimize database queries and indexing strategies for PostgreSQL, Microsoft SQL Server, and Oracle.
- Implement Redis caching to enhance application performance and reduce database load.

System Integration, API Development and Monitoring

- Deploy applications using Kubernetes (K8dash) and manage microservices at scale.
- Configure and optimize API Gateway for secure and efficient API management.
- Develop CI/CD pipelines for automated deployment and monitoring.

Security & Compliance

- Implement JWT authentication and OAuth2 for API security.
- Conduct security audits to identify and resolve vulnerabilities in APIs and backend services.
- Ensure compliance with industry security standards, including Log4j remediation.

Monitoring & Performance Optimization

- Utilize Kibana for centralized log management and real-time system monitoring.
- Conduct load testing and performance tuning to optimize API response times.
- Implement failover mechanisms and circuit breakers to improve system resilience.

Collaboration & Teamwork

- Work closely with frontend developers, DevOps engineers, and business stakeholders to deliver high-quality solutions.
- Maintain comprehensive technical documentation for system architecture, API specifications, and deployment processes.
- Support partner integration efforts, ensuring seamless communication between internal and external systems.
- Participate in technical discussions, code reviews, and knowledge-sharing sessions with user and ops team

Stack Highlights:

- **Languages:** Java, TIBCO
- **Frameworks:** Spring Boot, Spring Framework, TIBCO BW 5, TIBCO BWCE
- **Databases:** PostgreSQL, Microsoft SQL Server, Oracle
- **Version Control :** Github, Gitlab
- **Caching & Messaging:** Redis, Solace
- **Monitoring & Logging:** Kibana, K8dash
- **DevOps & Deployment:** Kubernetes, API Gateway, CI/CD Pipelines
- **Performance Testing:** K6 for load and stress testing, ensuring system scalability and reliability

Software Engineer specializing in backend development, system migration, and API management within the insurance industry. Experienced in large-scale data processing, microservices architecture, and enterprise system integration.

Experiences :

- Lead the migration of Flink-based services to Quarkus, improving system efficiency, maintainability, and reducing infrastructure costs.
- Develop and maintain RESTful APIs for e-KYC, contact detail updates, and policy services, ensuring high availability and scalability.
- Participate in the full software development lifecycle, including requirement analysis, design, implementation, testing, and deployment.
- Optimize ETL processes for SPAI transactions, leveraging Kafka as a message broker for real-time data streaming.
- Collaborate with cross-functional teams to maintain compliance with regulatory requirements and industry standards.

Selected Project Experience**PruServices**

- **e-KYC API** Provides identity verification services for policyholders by integrating with third-party KYC providers and internal customer databases. Implements security best practices, including encryption and OAuth2 authentication, to ensure data privacy and regulatory compliance.
- **Contact Detail Update Service** Manages real-time updates for customer contact details, integrating with core insurance systems and third-party communication services. Utilizes event-driven processing and message queues to ensure reliable data propagation across systems.

Pruhuh :

- **Migration Flink to Quarkus** Led the migration from Apache Flink to Quarkus to improve performance, reduce operational overhead, and enhance system maintainability. The migration included refactoring batch processing jobs, optimizing database interactions, and ensuring high system resilience during real-time data processing.
- **Withdrawal API** Developed and optimized an API for policyholder fund withdrawals, ensuring secure, efficient, and fast transaction processing. Integrated with banking services and internal financial systems, implementing robust fraud detection and validation mechanisms to prevent unauthorized transactions.
- **Switch Redirect** Implemented a switching and redirection service for policyholders managing their investment portfolios. Ensured real-time processing and compliance with regulatory policies, improving the user experience for financial transactions while maintaining high security and accuracy.

Job Desk & Responsibility**Backend Development & System Architecture:**

- Migrate and optimize backend services from Flink to Quarkus.
- Implement high-performance data storage solutions using Couchbase and PostgreSQL.
- Enhance and maintain Kafka-based messaging systems for real-time data processing.

Deployment & Server Management:

- Manage deployment pipelines using CI/CD tools.
- Optimize system performance by profiling APIs, database queries, and memory usage.

System Integration, API Development, and Monitoring:

- Develop and maintain microservices-based architectures.
- Integrate backend systems with IBM BPM and BRMS for process automation.
- Implement logging, monitoring, and alerting systems to ensure system reliability and availability.

Collaboration & Teamwork:

- Work closely with cross-functional teams to implement scalable solutions.
- Participate in technical discussions, code reviews, and knowledge-sharing sessions.

Stack Highlights:

- **Languages:** Java, Flink, Quarkus
- **Frameworks:** Spring Boot, Apache Kafka
- **Databases:** PostgreSQL, Couchbase
- **Version Control :** Bitbucket
- **DevOps & Deployment:** Kubernetes, API Gateway, CI/CD Pipelines
- **Business Rules & Workflow:** IBM BPM, BRMS

A backend software engineer specializing in designing and implementing scalable, high-performance financial systems. Experienced in System Architecture, API development, and optimizing backend processes to improve efficiency and security. Proficient in Spring Boot, Java, with a strong focus on System Integration, Authentication Mechanisms, And Performance Optimization

Experiences:

- Developed and designed backend systems for various MSIG Life projects, including the LINX and Andromeda teams.
- Focused on system design and backend development to enhance business operations.
- Built and integrated APIs to support financial and operational processes.
- Conducted system profiling and performance tuning for better application performance.
- Implemented authentication and security using OAuth2 and JWT.
- Developed CI/CD pipelines for automated deployments.
- Designed relational database schemas, optimizing indexes, foreign key constraints, and partitioning strategies for efficient query execution.
- Ensured data consistency, integrity, and security through database constraints and encryption mechanisms.
- Designed microservices architecture using Spring Cloud for service discovery, load balancing, and centralized configuration management.
- Implemented Kafka for event-driven processing and asynchronous communication between services.

Selected Project Experience**Fortify - Digital Life Insurance Platform (fortify.msiglife.co.id)**

A comprehensive life insurance platform that simplifies policy management, submission, and agent services. The system streamlines the end-to-end lifecycle of insurance policies, enabling better user experience and operational efficiency

Technical Contributions:

- Developed policy management and submission services using Spring Boot.
- Designed and implemented authentication services using Zuul Gateway, OAuth2, and JWT.
- Led migration of existing agent data into the Fortify system using Spring Batch, ensuring data consistency and integrity.
- Integrated various microservices using Spring Cloud and Kafka for seamless communication.
- Optimized performance by implementing caching strategies and database indexing in PostgreSQL and Microsoft SQL Server.

Andromeda - Digital Platform for Agency MSIG Life (Internal Web)

A digital platform designed to manage various agent-related operations, including Agent Alteration, Agent Compensation, Agent Debt, and Agent Financing.

Technical Contributions:

- Focused on developing the Agent Financing module, enabling financial services and loan processing for agents.
- Designed the database schema for the Agent Financing feature, ensuring efficient query execution and data integrity.
- Architected the microservices structure for the Agent Financing feature using Spring Cloud.
- Developed backend services to facilitate financing processes, including loan applications, approvals, and repayments.
- Designed and implemented the allowance calculation engine to determine financial benefits for agents based on predefined criteria.
- Optimized database queries and indexing to improve system performance.

Job Desk & Responsibilities**Backend Development & System Architecture:**

- Design and implement scalable microservices architecture using Spring Cloud.
- Utilize Eureka Server for service discovery and Zuul Gateway for API routing and security.
- Implement MyBatis for efficient database interaction with PostgreSQL
- Develop and maintain event-driven and asynchronous systems using Apache Kafka.
- Ensure system security and authentication using OAuth2, JWT, and API Gateway.

Database Design & Optimization:

- Design and optimize relational database schemas for PostgreSQL
- Implement indexing strategies, partitioning, and foreign key constraints to improve query execution.
- Ensure data consistency, integrity, and security through constraints and encryption mechanisms.
- Use MyBatis for efficient database access and query optimization.

Deployment & Server Management:

- Develop and maintain CI/CD pipelines for automated deployments.
- Manage deployments and configurations using Kubernetes and API Gateway.

System Integration, API Development, and Monitoring:

- Design, develop, and maintain RESTful APIs for internal and external integrations.
- Utilize Eureka Server for service discovery and load balancing.
- Implement logging, monitoring, and alerting systems to ensure high availability and reliability.

Collaboration & Teamwork:

- Work closely with cross-functional teams to implement scalable solutions.
- Participate in technical discussions, code reviews, and knowledge-sharing sessions.

Stack Highlights:

- **Languages:** Java
- **Frameworks:** Spring Boot, Spring Cloud, Apache Kafka, MyBatis
- **Microservices & API Gateway:** Eureka Server, Zuul Gateway
- **Databases:** PostgreSQL
- **Version Control:** Bitbucket
- **DevOps & Deployment:** Kubernetes, API Gateway, CI/CD Pipelines
- **Security & Authentication:** OAuth2, JWT, API Gateway

CERTIFICATION**DIGITAL TALENT KOMINFO : PROFESSIONAL ACADEMY***Digital Talent Scholarship : Data Science*

Jakarta, ID

2022**DIGITAL TALENT KOMINFO : PROFESSIONAL ACADEMY***Digital Talent Scholarship Programing : HTML, CSS and Javascript*

Jakarta, ID

2021**EDUCATION****UNIVERSITAS INDRAPASTA PGRI***Bachelor Degree Major in Computer Science; Minor in Computer Science*

Jakarta, ID

2022-Present**ENIGMA CAMP***IT Bootcamp (Full Stack Developer)*

Jakarta, ID

2020

A 3+ months intensive bootcamp learning Learn how to be an IT Developer and also other skill such as english communication, teamwork etc. provides 12 weeks hands-on coding curriculum that focuses on mobile development with react native. Web development with react js. Backend development with java, nodejs, and Golang. Database with MySQL, PostgreSQL, MongoDB

POLITEKNIK HARAPAN BERSAMA*Associate's Degree Major in Computer Science; Minor in Computer Engineering*

Tegal, ID

2016-2019**ADDITIONAL INFORMATION AND TECH STACK SUMMARY**

- **Program Management:** SDLC (Software Development Life Cycle), JIRA, Confluence
- **Languages:** Advanced in Java, JavaScript, SQL
- **Frameworks:** Spring Boot,,Quarkus, Flink, React Vue, NodeJS, Express
- **Message Broker :** Kafka, Solace
- **Caching :** Redis
- **Reporting :** Jasper Report, Ms. Excel
- **Microservices :** Eureka Server, Zuul Gateway
- **Testing :** K6, JMeter
- **Logging :** ELK Stack (Elastic Search, Logstash, Kibana), Promoteus, Grafana
- **Developer Tools:** Firebase, Tailwind, React Query, SASS, Testing Library, Babel, Webpack
- **Cloud & DevOps:** AWS, Kubernetes, Docker, Kafka
- **Databases:** PostgreSQL, Oracle, Couchbase
- **Version Control :** Github, Gitlab, Bitbucket

- **Relevant Skills:** Web Performance, Web Accessibility, Clean Code, Software Testing, CI/CD, REST API, GraphQL, JSON
- **Tools & Platforms:** MyBatis, IBM BPM, BRMS, PlantUML, Postman